
Multimodal AI for Precision Fitness: A Deterministic Pipeline for InBody-Driven Nutritional and Corrective Exercise Planning

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Abstract

Body composition data, such as InBody test results, provide critical information for precision fitness but are often underutilized by conventional applications that rely on limited metrics like BMI. This lack of personalization can result in stalled physical progress and increased injury risks due to unaddressed muscle imbalances. To address these challenges, we propose the InBody AI Fitness Assistant, a hyper-personalized multimodal AI system utilizing a deterministic, four-stage modular pipeline. The architecture begins with a Document Understanding Transformer (Donut) that extracts biometric matrices directly from InBody report images. The extracted data is then processed by a deterministic nutritional engine using the Katch-McArdle formula to calculate exact macronutrient targets based on the user's Lean Body Mass. Concurrently, a deterministic constraint-based filtering module identifies muscle weaknesses and queries a structured exercise dataset to recommend targeted, corrective gym movements. Finally, a zero-shot Large Language Model synthesizes these structured, exact outputs into an empathetic daily plan. By deliberately decoupling human-interpretable medical logic from generative functions, the proposed system minimizes the risk of AI hallucination, ensuring that recommendations remain medically precise, highly personalized, and safe for the user.

1 Introduction

1.1 Problem Statement

Body composition data, such as InBody test results, provide critical information for designing precision fitness programs [1]. However, many users struggle to translate these results into actionable plans. Most existing fitness applications still rely on traditional metrics such as Body Mass Index (BMI), which cannot distinguish between muscle mass and body fat [2]. As a result, body composition data are often underutilized, leaving users without meaningful guidance for improving their fitness.

1.2 Urgency & Impact

The inability to effectively interpret body composition data creates tangible physical risks. Ignoring muscle imbalances may increase the likelihood of biomechanical injuries, while static calorie targets can lead to stalled progress. Furthermore, the limited hyper-personalization of conventional fitness applications often results in users merely collecting health data without receiving training and nutrition guidance tailored to their individual needs [3].

1.3 Proposed Solution

To address these challenges, we propose the InBody AI Fitness Assistant, a hyper-personalized fitness assistant powered by Multimodal AI. The proposed system adopts a modular four-stage pipeline. In the first stage, Computer Vision, Optical Character Recognition (OCR), and Key Information Extraction (KIE) are employed to extract body composition metrics from InBody report images. The extracted data are then processed by deterministic rule-based algorithms to calculate personalized nutritional requirements. Next, a Machine Learning Recommender System identifies muscle weaknesses and recommends corrective exercises. Finally, a Large Language Model (LLM) with zero-shot prompting generates personalized coaching and explanations. By clearly separating deterministic calculations from the generative module, the proposed architecture minimizes AI hallucinations and ensures that every fitness plan remains precise, safe to follow, and grounded in validated numerical data.

2 Literature Review

2.1 Theories & Concept

2.1.1 Total Daily Energy Expenditure (TDEE)

Accurate nutritional guidance depends on precise estimation of Total Daily Energy Expenditure (TDEE) [4]. TDEE is calculated from the combination of Basal Metabolic Rate (BMR), the thermic effect of food, and energy expenditure through physical activity. In our proposed system, nutritional requirements are calculated using a rule-based approach based on the Mifflin-St Jeor equation, one of the most accurate methods for estimating daily calorie needs [5]. By separating deterministic calculations from the generative Large Language Model (LLM), the resulting calorie and nutritional targets remain precise and are not affected by AI-generated estimation errors.

2.1.2 Key Information Extraction (KIE) for Document Processing

Key Information Extraction (KIE) is a Computer Vision technique that identifies and extracts specific information from images or documents [6]. To extract numerical measurements from InBody report images, modern KIE systems go beyond conventional Optical Character Recognition (OCR) by integrating Deep Learning models capable of understanding the visual layout and document structure as a whole [6]. In addition, synthetic data are used during model training to improve robustness, enabling the system to accurately recognize InBody measurements even when uploaded images are tilted, shadowed, or slightly blurred.

2.2 Existing Solutions and Limitations

Most fitness applications currently available still rely on generic template-based approaches. These applications typically use only basic user information, such as age, body weight, and estimated exercise intensity. Since they cannot continuously adapt training recommendations to users' physical progress, maintaining long-term user engagement remains a significant challenge [7].

In contrast, the InBody AI Fitness Assistant provides a hyper-personalized approach by analyzing body composition in much greater detail. While conventional applications often overlook muscle imbalances, such as strength differences between the upper and lower body that may increase injury risk, our system addresses this gap through a Content-Based Filtering approach [8]. The recommender system automatically identifies muscle weaknesses from InBody measurements and proactively suggests appropriate corrective gym exercises. This transforms passive body composition data into personalized workout recommendations, following the concept of sports recommender systems that leverage individual user characteristics to deliver more relevant recommendations.

3 Methodology

This section details the technical foundation of the proposed AI fitness assistant. Given the critical nature of medical accuracy and user safety, our approach deliberately avoids end-to-end generative reliance. Instead, we design a deterministic, multimodal architecture that keeps

human-interpretable logic in control at every stage. The following subsections walk through our four-stage pipeline, spanning computer vision, rule-based medical logic, constraint-based exercise filtering, and natural language generation, alongside our data strategy for maintaining privacy compliance and evaluation rigor.

3.1 AI/ML Pipeline Design

To ensure computational efficiency, deterministic medical safety, and high personalization, the proposed solution employs a four-stage modular architecture. This separation of concerns strictly prevents the hallucination risks commonly associated with Large Language Models (LLMs) in health-critical applications.

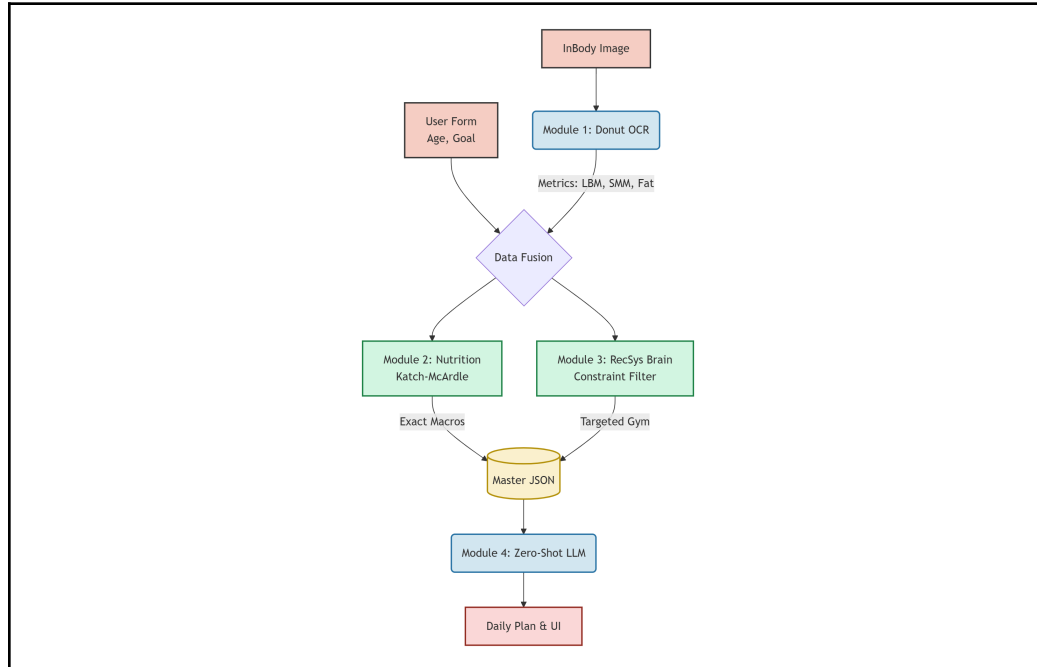


Figure 1: Proposed Pipeline Flow

3.1.1 User Onboarding & Visual Extraction Module

The workflow initiates with a brief user onboarding form to capture baseline metrics: Age, Biological Sex, Activity Multiplier (e.g., sedentary vs. active), and Fitness Goal (hypertrophy vs. fat loss). Subsequently, the user uploads an InBody test sheet. We utilize the Donut (Document Understanding Transformer) architecture, which bypasses traditional OCR pipelines by directly mapping the document's visual layout. This model extracts a highly structured JSON payload containing the Basal Metabolic Rate (BMR), Percent Body Fat (PBF), Skeletal Muscle Mass (SMM), Visceral Fat Level, and Segmental Lean Analysis.

3.1.2 Deterministic Nutritional Engine

The combined data (User Form + InBody JSON) is routed to a rule-based Python logic engine. To maximize the utility of the extracted InBody data, this engine applies the Katch-McArdle Formula, calculating the Total Daily Energy Expenditure (TDEE) based specifically on the user's Lean Body Mass rather than generic total weight. The engine deterministically distributes this caloric target into strict macronutrient and micronutrient thresholds (exact grams of Protein, Carbohydrates, Fats, and Fiber) tailored directly to the user's stated Fitness Goal.

3.1.3 Deterministic Constraint-Based Filtering (Exercise Recommendation)

Serving as the core analytical brain, this module replaces generic predictive modeling with a structured querying system against an exercise database, strictly guided by the user profile and InBody metrics.

3.1.3.1 Imbalance Correction

By evaluating the Segmental Lean Analysis, an asymmetry exceeding a defined threshold (e.g., >5% bilateral SMM deviation between left and right limbs) triggers strict rules to prioritize targeted unilateral corrective movements (e.g., *Bulgarian Split Squats*).

3.1.3.2 Goal Driven Adaptation

The system modulates the exercise selection based on the onboarding Fitness Goal. Hypertrophy goals dynamically adjust the filter to prioritize progressive overload on bilateral compounds, whereas fat loss goals combined with high Visceral Fat readings increase the frequency of cardiovascular and HIIT parameters within the output JSON.

3.1.4 Natural Language Generation (NLG) Interface

The deterministic, mathematically sound outputs from Modules 2 and 3 are concatenated into a finalized Master JSON string. This strict payload is injected into an OpenAI LLM via API using zero-shot prompting. The LLM functions strictly as a linguistic synthesizer, transforming the rigid dataset into an empathetic, human-like progress report and daily plan without altering the underlying programmatic logic.

3.2 Planned Use of Data

The development and operation of this system rely on a hybrid data strategy, carefully designed to bypass medical privacy constraints while maintaining high accuracy:

3.2.1 Synthetic Data for OCR

Due to strict health data privacy regulations, acquiring real InBody test sheets is unfeasible. To train the visual extraction model, we will utilize Synthetic Image Generation. A Python-based script will algorithmically generate thousands of artificial InBody templates with randomized numerical values and dates. To ensure model robustness against real-world smartphone capture conditions, visual augmentations such as Gaussian blur, varying lighting gradients, and spatial distortion will be applied to the synthetic dataset.

3.2.2 Exercise Recommendations Dataset

The recommendation engine (Module 3) will be grounded in the open-source Fitness Exercises Dataset (sourced from <https://github.com/hasaneyldrm/exercises-dataset>). This JSON-based dataset provides a comprehensive taxonomy of physical exercises, categorized rigorously by target body parts, secondary muscles, and required equipment. This structured taxonomy allows the algorithm to precisely query and map corrective unilateral or bilateral movements to the user's specific muscular imbalances detected in the OCR phase.

3.2.3 Nutritional Standards and Validation

The nutritional logic (Module 2) does not require machine learning training data. Instead, it relies on deterministic medical formulas as its baseline truth. To extend this feature in the future to include specific food recommendations, the system is designed to integrate seamlessly with the USDA FoodData Central database. This ensures that every macronutrient calculation mapped to a real-world food item is medically sound and statistically verified.

3.3 Modeling Approach

The design philosophy guiding our system architecture, illustrated in Figure 1, centers on maximizing user safety by confining generative mechanisms to linguistic tasks while keeping logical calculus deterministic. We employ a hybrid multimodal pipeline consisting of four distinct modules.

Module 1 utilizes a computer vision-based Document Understanding Transformer (Donut) [9]. The architecture is fine-tuned using standard Cross-Entropy Loss applied to next-token predictions of the target JSON. To achieve robust generalization against smartphone camera distortions, the model is trained on approximately 5,000 synthetic InBody sheets, evenly distributed between InBody 270 and InBody 570 layouts (2,500 samples each). These samples are heavily augmented with Gaussian blur, rotation, and lighting distortions to extract biometric matrices with high fidelity, mapping visual document layouts directly to structured JavaScript Object Notation (JSON) payloads without a traditional OCR step.

The logical core follows a decoupled rule-based approach. The visual payload (e.g., Lean Body Mass (LBM)) and user form inputs are fused and processed by a deterministic nutritional engine (Module 2) employing the Katch-McArdle formula, ensuring physiologically precise macronutrient calculations derived from dynamic LBM. Concurrently, a gym recommendation system (Module 3) employs Deterministic Constraint-Based Filtering rather than computationally intensive vector techniques (e.g., Dense Embeddings), which carry a risk of hallucination. Module 3 directly leverages the structured JSON attributes (target, bodyPart, equipment) from the *hasaneyldrm/exercises-dataset* repository. When muscle asymmetry is detected, it executes Rule-Based Boolean Querying via exact-match logic to deterministically prioritize safe, corrective movements.

The outputs from Module 2 and 3 are consolidated into a structured Master JSON. This intermediate state serves as the input constrained context for the zero-shot Large Language Model (Module 4). To strictly prevent the LLM from mutating medical and caloric calculations, we implement a dual-validation mechanism. At the API level, we utilize OpenAI's Structured Outputs configured with a strict JSON Schema. Subsequently, at the backend level, the LLM's JSON output undergoes Pydantic validation. If Pydantic detects any numerical mutation from Module 2's outputs, the system triggers a validation error, drops the generative response, and automatically defaults to deterministic fallback text.

3.4 Future Development and Scalability

The proposed solution is designed as a preliminary proof-of-concept; however, its modularity provides robust vectors for horizontal scaling and enhanced accuracy in future iterations:

3.4.1 Modular Scalability and Edge Computing Deployment: As each component of the pipeline is decoupled by structured I/O (JSON), Module 1 (OCR) can be iteratively upgraded or shifted to a lightweight architecture for on-device edge AI deployment (e.g., using quantization). This reduces latency and improves data privacy by keeping image processing localized. Module 4 can likewise be updated to a newer or domain-specific LLM without rebuilding the deterministic logical engines.

3.4.2 Multimodal Progression Tracking and Form Analysis: Future versions can integrate Computer Vision-based human pose estimation via a device's camera for real-time exercise form feedback, adding a "coaching" layer to the "planning" layer. Longitudinal tracking will also be added, analyzing sequential InBody inputs to dynamically adapt calorie recommendations using algorithms to predict dynamic TDEE shifts.

3.4.3 Enhanced Medical Compliance and Data Privacy: While currently using synthetic data and publicly available datasets to maintain medical privacy compliance during Datathon development, future iterations will focus on integrating with secure, HIPAA- or GDPR-compliant personal health information (PHI) infrastructure to handle actual patient data and ensure complete end-to-end data auditing.

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